Leveraging Permuted Image Restoration for Improved Interpretation of Remote Sensing Images

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***Abstract*— In this study, we introduce a novel self-supervised learning adapter based on permutated image restoration (PIR) for effectively transferring pretrained weights from natural images to remote sensing object detection tasks. The adapter's unique methodology encompasses a three-phase process: segmenting and permuting image blocks, estimating permutation matrices for sequence reconstruction, and applying specialized loss functions for accurate block positioning. The use of our approach results in the maintenance of fidelity in both absolute and relative block positions as demonstrated by the evaluation of block similarities. The empirical results indicate significant performance enhancements for diverse datasets spanning optical and synthetic aperture radar data types, including high resolution ship collections 2016 (HRSC2016), Small Object Detection dAtasets - Aerial (SODA-A), and rotated ship detection dataset (RSDD) while effectively avoiding overfitting.**

***Index Terms*— Oriented object detection, permutated image restoration (PIR), permutation matrix estimation, pretrained weights, remote sensing, self-supervised learning.**

I. Introduction

THE emergence of deep learning has heralded significant advancements in the intelligent interpretation of remote sensing data [1], [2], [3]. However, a critical challenge persists due to the substantial disparity between natural and remote sensing images. The difficulty in generalizing pretrained weights from natural image datasets, like ImageNet, to remote sensing data remains a significant hurdle [4], [5], [6]. This scenario underscores the imperative need to develop pretrained weights specifically tailored for remote sensing images, with the aim of augmenting their intelligent processing capabilities.

Obtaining expansive remote sensing datasets is arduous, impeded by the complexities of image acquisition and annotation, particularly in data-scarce application scenarios. While the use of traditional training models is sufficient in data-abundant environments, they often result in overfitting in the context of limited datasets. This challenge necessitates the adoption of a label-independent self-supervised learning approach that is focused on extracting intrinsic data features, thereby increasing the adaptability of pretrained weights to remote sensing imagery.

When applying established self-supervised methods from natural images to remote sensing, especially synthetic aperture radar (SAR) images, formidable challenges are encountered [7]. For instance, the masked prediction strategy [8] struggles with remote sensing images, as the strong correlation between masked and unmasked sections found in natural imagery is often lacking [9]. An example is illustrated in Fig. 1(a), where masking a ship against an ocean backdrop may not yield meaningful insights for the model. Contrastive learning, while effective for natural images, faces unique obstacles when applied to remote sensing images [10], [11]. For example, as depicted in Fig. 1(b), after undergoing data augmentations like rotation and random cropping, the different images may appear too similar, impeding the network in learning from positive and negative pairings.

We propose, as a solution, a self-supervised method based on permuted image restoration (PIR), specifically designed for remote sensing images. As detailed in Fig. 2, our method involves a three-step process. The network processes the permuted image and predicts a permutation matrix, in alignment with the permutation process. The loss functions target a random subset of the image blocks, focusing on both their absolute and relative positions, determined by comparing block similarities in the original and predicted images.

The effectiveness of our approach is evidenced through extensive experiments on diverse remote sensing image datasets, including high resolution ship collections 2016 (HRSC2016) [12], Small Object Detection dAtasets - Aerial (SODA-A) [13], and rotated ship detection dataset (RSDD) [14], targeting the challenging task of oriented object detection. We observed substantial improvements in performance, with the SODA-A dataset experiencing a 0.6 increase, HRSC2016 a 0.5 uplift in the mAP(07) metric, and notably, the RSDD dataset exhibiting a 1.4 improvement in detecting inshore objects. Furthermore, tests on raw Gaofen-3 imagery confirmed the ability of our method to enhance the performance of object detection under varied imaging conditions by leveraging the spatial context inherent in remote sensing images. Our contributions are threefold: First, we propose a unique self-supervised approach that utilizes the inherent sequence of image blocks as a natural pseudolabel, representing a distinct departure from existing methods. Second,

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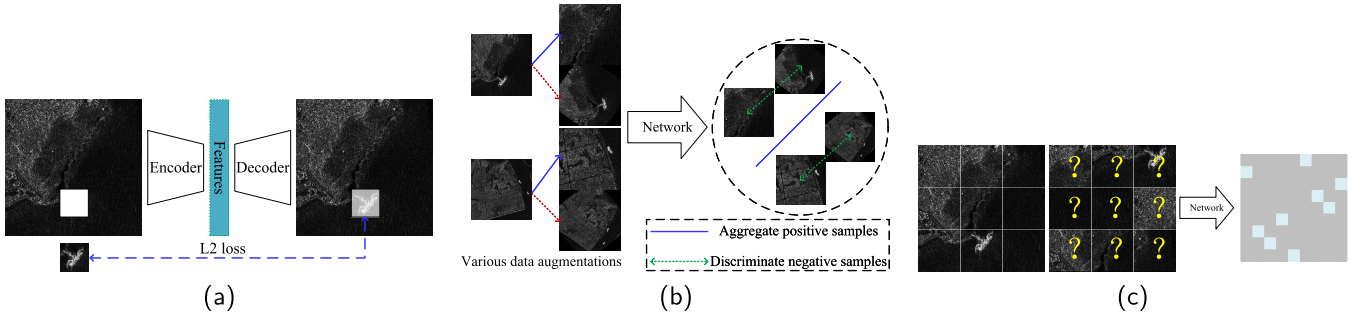


Fig. 1. Comparisons of different self-supervised learning methods applied to remote sensing images. (a) Masked prediction from a masked image, where parts of the image are randomly masked and then predicted. (b) Contrastive learning, which distinguishes between positive and negative examples obtained from the same image based on different data augmentations. (c) Position prediction from a permuted image, where the image is divided into blocks, randomly shuffled and then restored by predicting the original order. For enhanced clarity and details, consider enlarging the image.

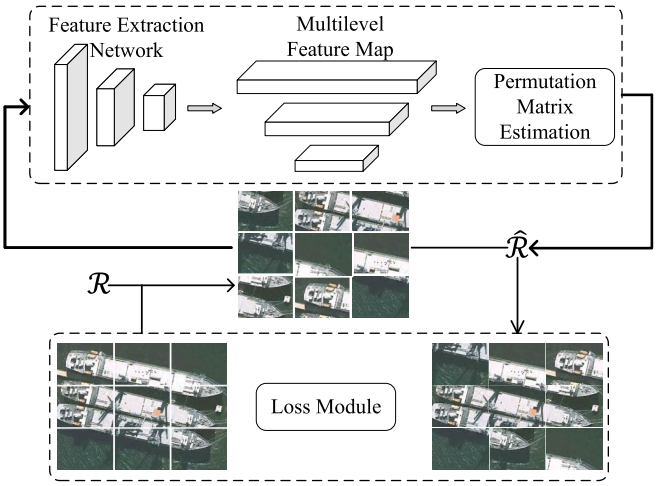


Fig. 2. Overview of the proposed network architecture, which includes a three-stage procedure for PIR. In the first stage, the image is divided into blocks and then randomly shuffled to produce a permuted image. In the second stage, the permuted image is passed through the network to predict a permutation matrix corresponding to the applied permutation process. In the third stage, loss functions are applied to a randomly selected subset of the blocks in the restored image. Both absolute positions and relative positions are considered, determined by comparing the similarity between the four adjacent blocks in the original image and in the predicted image.

we adapt this approach to remote sensing, customizing pretrained weights from natural images to meet the specific needs of remote sensing imagery. Finally, through a comprehensive ablation study, we demonstrate the effectiveness of our method across various datasets.

The rest of this article is structured as follows: related work is reviewed in Section II. Delineated in Section III is our proposed methodology, including image permutation preprocessing, permutation matrix estimation, and the formulation of a dedicated loss function. Presented in Section IV is the empirical validation of our approach, focusing on experiments conducted on the HRSC2016, SODA-A, and RSDD datasets for oriented object detection. Finally, the key contributions of the study are summarized in Section V, with a discussion of the limitations of the proposed method.

II. Related Work

A. Self-Supervised Learning and Transfer Learning

Self-supervised learning, distinct from supervised learning, exploits unlabeled data, thus reducing the dependency on costly manual annotation. Its ability to leverage abundant, freely available data results in models with enhanced generalization and adaptability to data distribution shifts.

The key methods of self-supervised learning, such as contrastive learning [10] and masked prediction [8], have garnered significant attention. Contrastive learning optimizes model representations by distinguishing semantically similar and dissimilar instances, thereby enhancing feature uniqueness. Meanwhile, masked prediction methods compel models to infer hidden segments of input data, thereby fostering deeper structural and contextual understanding. These approaches are exemplified by Bidirectional Encoder Representations from Transformers (BERT) [15] in language processing and various imaging and painting techniques.

In transfer learning [16], knowledge is repurposed from one domain to enhance performance in another related but distinct domain. Being particularly beneficial in low-data scenarios, it leverages pretrained models from data-rich domains to boost effectiveness in target tasks. This underscores the importance of creating impactful pretext tasks in self-supervised learning, which is essential for deriving valuable representations.

B. Jigsaw Puzzle Solving as a Pretext Task

Using the jigsaw puzzle as a self-supervised pretext task has become an innovative approach to improving perceptual understanding of image content by predicting the spatial relationships between blocks [17]. However, most existing methods involve parallel processing of different image blocks fed into the network, resulting in limited feature-level interaction, and thus, insufficient learning [18]. The limitations regarding the output of these methods generally revolve around three perspectives: possible permutations [19], absolute position prediction [20], and relative position prediction [21]. For example, considering the probability space of possible permutations is a widely used approach. However, it is limited by the enormous number of possible combinations, which results in a very large number of prediction categories for the network. If only a portion of the probability space is sampled, this can lead to a loss of rich learning information such that the potential of this pretext task is not fully exploited, as shown in Fig. 3(c). Absolute position prediction is another commonly used strategy. As shown in Fig. 3(a), this technique aims to directly determine the position of each block. However, a major drawback of this approach is that it can lead to similar

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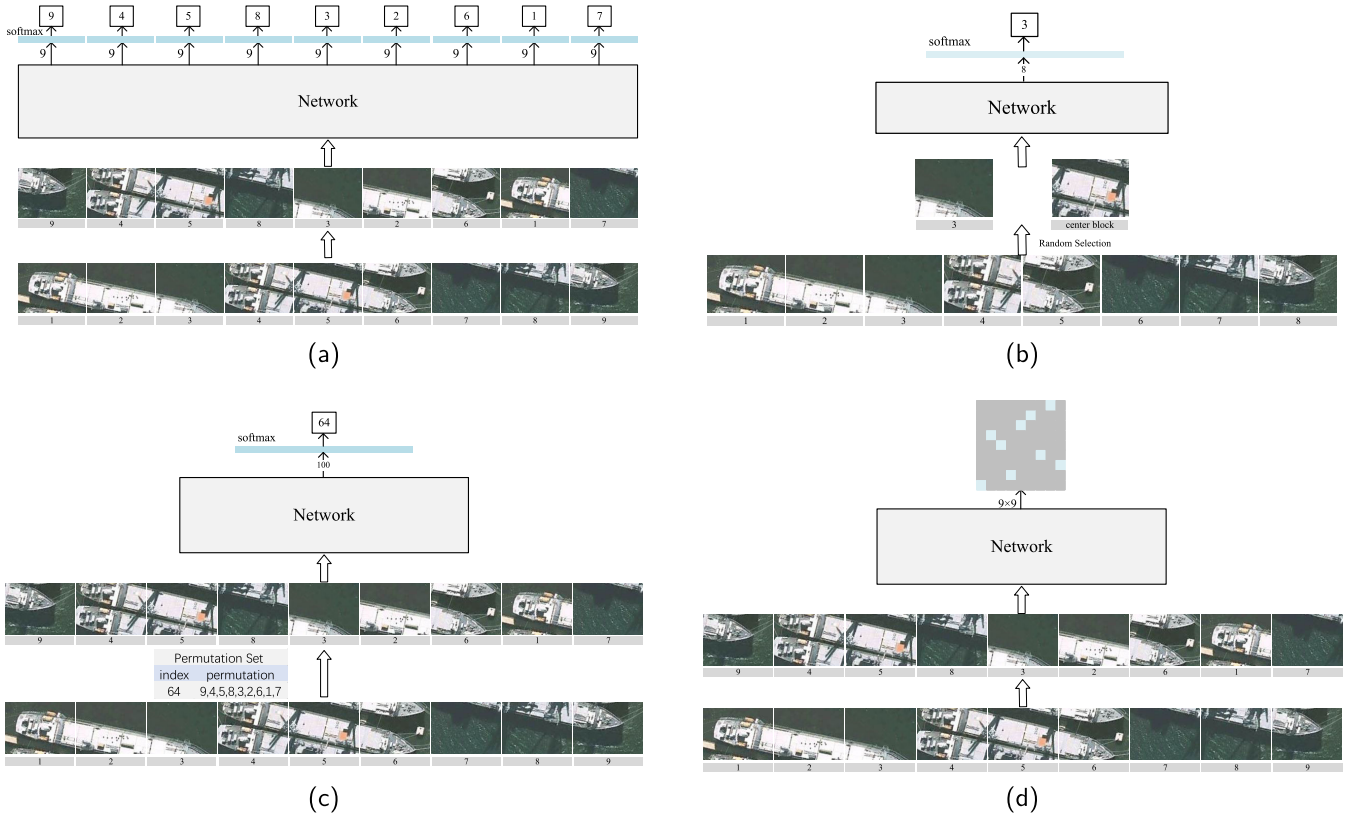


Fig. 3. Comparisons of different strategies for predicting the order of image blocks. (a) Predicting the absolute position of each block. (b) Predicting the relative position of any adjacent block to the center block. (c) Predicting which of the 100 predetermined types of permutation combinations the current method refers to. (d) Applying our proposed method to directly predict the permutation matrix corresponding to the permutation process (with the maximum value in each row in a unique column). For enhanced clarity and details, consider enlarging the image.

blocks inadvertently predicting the same position, thus limiting the network's ability to effectively differentiate. Moreover, deriving an absolute position for a single block (while considering the other blocks as a monotone background) seems rather illogical. Relative position prediction is another method. While it provides a more nuanced perspective by taking into account the positional relationships between blocks, it suffers when it comes to a holistic understanding of the image, especially on a global scale, as shown in Fig. 3(b). Comparing the feed of paired blocks to the network with the feed of all image blocks, we find that the network's learning data source is significantly limited in the case of the former method.

Despite their limitations, some researchers have attempted to combine these methods of relative and absolute position predictions [22]. However, it has proven difficult for networks to truly learn from the jigsaw puzzle task, due to its inherent challenges. To overcome these obstacles, some researchers have explored self-supervised multitask learning, combining the puzzle task with classification tasks [20]. However, despite some advances, these methods fail to fully exploit the rich spatial contextual information inherent in images. The limitations of each approach, such as the focus on specific block segmentation strategies, the limited perception of the entire image, and the inability to fully account for the probability space, highlight the need for new techniques. Ideally, these techniques should consider both absolute and relative positions and also address the potential problem of identical positions being predicted by the network. The ultimate goal should be to use as much spatial contextual information about the image as possible to maximize the learning potential.

C. Oriented Object Detection

Due to the fact that objects in remote sensing images often have irregular geometric shapes, conducting horizontal bounding box detection would result in a large number of redundant regions. Oriented object detection is a downstream task that has attracted considerable attention in remote sensing image understanding. Recently, Single-shot Alignment Network (S2ANet) [23], oriented Regions with CNN features (RCNN) [24], Region of Interests (RoI) transformer [25], and convex-hull feature adaptation (CFA) [26] have achieved outstanding performance in oriented object detection in remote sensing images by leveraging advanced experiences in horizontal bounding box detection. In addition, works on Gaussian Wasserstein distance (GWD) [27] and Kullback-Leibler Divergence (KLD) [28] are examples of where the network structure design was optimized from the perspective of loss function design. However, despite the significant progress of oriented object detection in remote sensing image understanding, a few works have studied the impact of pretrained models on downstream tasks in remote sensing image understanding. In the aforementioned works, weights obtained from representation learning on natural images are still being used for pretraining models, which can lead to suboptimal convergence results and slow convergence speed. In this article, we design an unsupervised learning approach tailored to the characteristics of remote sensing images to benefit downstream tasks.

III. Methodology

The methodology of this work is explained in Sections III-A–III-C and includes three main procedures: image

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permutation preprocessing, permutation matrix estimation, and loss function design. In the image permutation preprocessing phase, the input images are first segmented into individual blocks and then randomly shuffled. The shuffled blocks are then reassembled into a permuted image, which is then fed into the feature extraction network for further analysis. Through this process, the pretext task of permutated image restoration (PIR) can be decoupled from the feature extraction network and the image block partitioning strategy such that it can be applied to different feature extraction networks. Moreover, the difficulty of the pretext task can be dynamically adjusted according to the progress of network learning.

The whole process of image block shuffling can be viewed as a left multiplication of the input image block sequence with a permutation matrix (each row and column contains only one "1," and the rest are "0"s). The pseudolabel for the pretext task to recover the image mixture is this matrix. Consequently, in the permutation matrix estimation phase, we segment the feature map output from the feature extraction network using the same block slicing strategy and predict a permutation matrix estimate based on these feature block sequences.

Then, we define the loss function based on the estimated permutation matrix. Each feature block has its relative and absolute position. We developed a neighborhood-based loss function for the relative position and constrained the absolute position of the image blocks using a cross-entropy (CE) loss function. Considering that imposing position constraints on all image blocks might cause the target task to become too strict and make it difficult for the network to converge to the optimal solution, in this work, constraints on the loss function are imposed only on a randomly selected portion of the image blocks. The aim of this strategy is to ensure prediction accuracy while avoiding excessive constraints.

A. Image Permutation Preprocessing

In previous studies, the input images were segmented into blocks that were simultaneously into the same feature extraction network [19], [21], [22], [29]. After interacting with the corresponding feature blocks, these networks output the predicted location information. The limitations of this approach are evident in two key areas. First, the interaction of features between image blocks is insufficient, lacking depth and complexity. Second, the need to develop specific interaction methods for feature blocks according to the particular network architecture limits the universality of the pretext task.

In this study, we present a new method in which the image is segmented, permuted, and reassembled prior to network input. Preprocessing in this way does not change the shape of the original network input and only randomizes the positions of the image blocks. This approach allows the pretext task to be decoupled from the specific network and increases the universality of the task. Moreover, this method does not require additional trainable network parameters. Consequently, it provides flexibility in dynamically adjusting the image block segmentation strategy during training, which significantly increases the adaptability of the network.

Specifically, the procedure begins by splitting the input image, denoted P, into a set of subimage blocks. This set, denoted as $\boldsymbol{\mathcal{P}}$, is arranged in a grid format of m rows and n columns. Each subimage block is assigned an identifier, arranged from top to bottom and left to right. Thus, the ith subimage block is expressed as $\mathcal{P}_i$, where i belongs to the integer series $\{0, 1, \ldots, mn-1\}$. Then, a permuted sequence $\boldsymbol{R}$ is randomly generated. The sequence is defined as $[R_0, \ldots, R_i, \ldots, R_{mn-1}]$, where each element R_i comes from the integer set $\{0, 1, \ldots, mn-1\}$. Exploiting this permuted sequence $\boldsymbol{R}$, a new sequence of subimage blocks, represented as $\tilde{\boldsymbol{\mathcal{P}}}$, is assembled. This sequence adheres to the order given by $\boldsymbol{R}$ such that $\tilde{\mathcal{P}}_i = \mathcal{P}_{R_i}$ for each i in the integer range $[0, mn)$. After creating the reordered subimage block set $\tilde{\boldsymbol{\mathcal{P}}}$, a recomposed image, denoted $\tilde{P}$, is generated. The resulting image $\tilde{P}$ differs from the original image P only with respect to the reordered subimage blocks. Moreover, it is worth noting that the permuted sequence $\boldsymbol{R}$ can be conceived as a permutation matrix denoted $\boldsymbol{\mathcal{R}}$. Therefore, the permutation process can be viewed as a left multiplication of the original sequence of subimage blocks with a permutation matrix, which can be formally represented as follows:

$$\begin{pmatrix} \tilde{\mathcal{P}}_0 \\ \vdots \\ \tilde{\mathcal{P}}_i \\ \vdots \\ \tilde{\mathcal{P}}_{mn-1} \end{pmatrix} = \begin{pmatrix} \mathcal{P}_{R_0} \\ \vdots \\ \mathcal{P}_{R_i} \\ \vdots \\ \mathcal{P}_{R_{mn-1}} \end{pmatrix} = \boldsymbol{\mathcal{R}} \times \begin{pmatrix} \mathcal{P}_0 \\ \vdots \\ \mathcal{P}_i \\ \vdots \\ \mathcal{P}_{mn-1} \end{pmatrix} \tag{1}$$

where $\mathcal{R}_{i,j}$ is defined as follows:

$$\mathcal{R}_{i,j} = \begin{cases} 1, & \text{if } j = R_i, \\ 0, & \text{otherwise.} \end{cases} \tag{2}$$

B. Permutation Matrix Estimation

This section is devoted to analyzing and addressing the key challenge in self-supervised learning for permuted image recovery: determining a logical output for the positions of the different image blocks. Given a strategy of splitting images into an m-row and n-column grid, there are $(mn)!$ possible permutations and each image block has mn potential positions. The existing research methods can generally be classified according to three main directions. In the first approach, the problem is considered from the entire probability space [19], allowing the network to output probability values for different possibilities. However, the size of $(mn)!$ can be immense, leading to a strong imbalance of positive and negative samples. For example, if $m = n = 3$, there are $9! = 362\,880$ possibilities. Therefore, Noroozi and Favaro [19] used a greedy algorithm to select 100 possibilities out of 9! and, during the partitioning process, select a random block-cutting strategy from these 100 possibilities. The network then outputs probability values for these 100 partitioning strategies. With this method, however, learning is restricted to a specific set of partitioning strategies, thus limiting the scope of learning. In the second approach, the absolute position of each block is independently considered [29]. Since each block has mn possible positions, the researchers directly output a fixed dimension of probabilities for different positions after passing each block feature through a fully connected layer. However, this method cannot constrain the relative positional relationships between different blocks, which may result in

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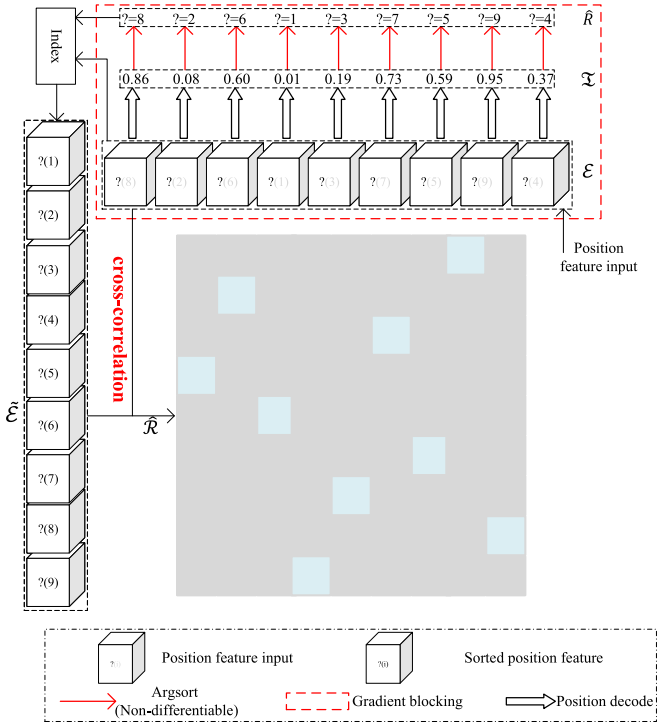


Fig. 4. Use of positional correlation equivalence as an alternative to the nondifferentiable argsort operation. The red dashed box highlights an intuitive approach where each positional feature vector outputs a value to denote its position, which is then sorted. However, the nondifferentiability of the argsort operation hinders efficient network training. Our proposed method overcomes this hurdle by employing a correlation-based strategy that uses the original input sequence and the sorted sequence to obtain an equivalent permutation matrix.

learning features that lack contextual associations. In the third approach, the relative positional relationships between different blocks are considered. Doersch et al. [21] attempt to predict the positions of the eight neighboring blocks relative to the central block. However, this approach has its limitations. It provides a limited number of features that the network can learn, and if an image has similar blocks at different positions, the concept of position may become ambiguous, affecting the network's ability to make reasonable predictions. In addition, Wei et al. [22] attempted to integrate these three perspectives by constraining the network predictions using multiple loss functions. However, this method is very complex and experiences difficulty in effectively accounting for all aspects.

The methods described earlier do not fully exploit the rich spatial contextual information in the images, which limits the possibilities of network learning. In our work, we attempt to address this problem from a new perspective. We assume that the pseudolabels for the pretext task of permuted image recovery are actually the reordering sequence $\boldsymbol{R}$ or the permutation matrix $\boldsymbol{\mathcal{R}}$ (both are mathematically equivalent), as shown in Section III-A. Can we instruct the network to predict the permuted sequence R? Considering the positional sequence necessarily involves comparing values, which ultimately requires that each feature block be converted into a value representing the positional sequence. The absolute value of this would indicate the position of the corresponding image block. The initial intuition would be to sort these values representing the positions of the image blocks to obtain values of their position indices. However, this process is clearly not differentiable. This nondifferentiability explains why this perspective has not been considered in previous research. Since a direct output of R by the network is not feasible (since gradients cannot be transmitted backward and network parameters cannot be effectively trained), we propose to let the network output an estimated value of the permutation matrix $\boldsymbol{\mathcal{R}}$. This entire matrix consists of $(mn)^2$ values, where each row holds its maximum value at a unique column. Previous studies [29] have predominantly focused on individual rows and output the probability values of each block at varying positions. However, these approaches lacked an effective mechanism to constrain the columns, which could lead to different image blocks predicting identical positions. In this study, we output the estimated value of the permutation matrix $\boldsymbol{\mathcal{R}}$ from the viewpoint of self-correlation.

Illustrated in Fig. 4 is a clever method for circumventing the nondifferentiability of the argsort operation, a common problem in processing position feature vectors. As shown in the red dashed box, there are values representing the position of each position vector, which are subsequently ordered by sorting. However, the nondifferentiability of this argsort operation is an impediment to efficient network training. Our solution to this dilemma is to implement a correlation-based technique to obtain an equivalent permutation matrix. In this alternative, the focus is solely on mathematically extracting values representing the position from the position vectors without including any training parameters. With this method, we return to the core of recovering permuted images and focus primarily on obtaining a value that could represent the position. Our interest lies in the relative magnitude of the output values given by the different blocks, which better corresponds to the intrinsic features of position precedence. In previous work, position encoding via transformer-based networks has been introduced for providing a mapping scheme from position to feature. A key question we want to address is: How can we guide the network to logically extrapolate the position from the feature?

Take a randomly permuted image $\tilde{P}$, which is passed through an arbitrary feature extraction network to obtain the corresponding set of feature maps, $\boldsymbol{F}$. The same partitioning strategy as for the input image is applied to these feature maps, $\boldsymbol{F}^k$, individually partitioning each into a combination of m rows by n columns of subfeature blocks, denoted $\boldsymbol{\mathcal{F}}^k$. The subfeature blocks are then numbered from left to right and from top to bottom to obtain $\boldsymbol{\mathcal{F}}_i^k$, where $i \in \{0, 1, \ldots, mn-1\}$. The positional feature vector $\boldsymbol{\mathcal{E}}_i^k$ of the subfeature block $\boldsymbol{\mathcal{F}}_i^k$ can be extracted using the following:

$$\boldsymbol{\mathcal{E}}_i^k = \sigma\left(\mathrm{MLP}\left(P_a\left(\boldsymbol{\mathcal{F}}_i^k\right)\right) + \mathrm{MLP}\left(P_m\left(\boldsymbol{\mathcal{F}}_i^k\right)\right)\right) \tag{3}$$

where P_a and P_m represent the average pooling and max pooling operations, respectively, and σ represents the sigmoid activation function. Here, MLP stands for multilayer perceptron. According to our previous analysis, it is necessary that each subfeature block outputs a value that can represent its positional order. The pooling operation in (3) allows the output of positional feature vectors with a uniform shape, regardless of the size of the feature block. There are numerous methods to derive a representative value for the position from the positional feature vector, including direct methods that

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utilize average, maximum, or minimum values. In this work, we propose a new approach based on inverse position decoding (IPD). Given an input sequence of length mn, we denote i as its position within the sequence and $\boldsymbol{p_i} \in \mathbb{R}^d$ as the positional feature vector corresponding to position i, where d denotes the dimension of the vector. The function $f : \mathbb{N} \to \mathbb{R}^d$ is established to generate the positional feature vector, adhering to the formula

$$p_i^{(t)} = f(i)^{(t)} = \begin{cases} \sin(\omega_r \cdot i), & \text{if } t = 2r \\ \cos(\omega_r \cdot i), & \text{if } t = 2r+1 \end{cases} \tag{4}$$

where the frequency ω_r is defined as follows:

$$\omega_r = \frac{1}{1000^{2r/d}}. \tag{5}$$

From the perspective of an approximate inverse function, we can derive an approximate estimate of the position index i from $\boldsymbol{\mathcal{E}_i^k}$. $\mathbb{F} : \mathbb{R}^d \to \mathbb{N}$, a function that produces a representative position value, $\mathfrak{T}_i^k$, from the positional feature vector $\boldsymbol{\mathcal{E}_i^k}$, which can be defined as follows:

$$\mathfrak{T}_i^k = \mathbb{F}\left(\boldsymbol{\mathcal{E}_i^k}\right) = \frac{1}{d}\sum_t \begin{cases} \dfrac{\arcsin\left(\mathcal{E}_i^{k(t)}\right)}{\omega_r}, & \text{if } t = 2r \\ \dfrac{\arccos\left(\mathcal{E}_i^{k(t)}\right)}{\omega_r}, & \text{if } t = 2r+1. \end{cases} \tag{6}$$

To estimate the permuted sequence $\hat{\boldsymbol{R}}^{\boldsymbol{k}}$, we perform an argsort operation on the set $\boldsymbol{\mathfrak{T}^k}$ that orders the elements in ascending order and returns the corresponding indices. Formally, for each $i \in [0, mn)$, we define $O(\mathfrak{T}_i^k, \boldsymbol{\mathfrak{T}^k}) = \text{argsort}(\mathfrak{T}_i^k, \boldsymbol{\mathfrak{T}^k})$, which means that $\mathfrak{T}_i^k$ is the $O(\mathfrak{T}_i^k, \boldsymbol{\mathfrak{T}^k})$-smallest element of the set $\boldsymbol{\mathfrak{T}^k}$. Therefore, $\hat{\boldsymbol{R}}^{\boldsymbol{k}}$ can be represented as $\hat{\boldsymbol{R}}^{\boldsymbol{k}} = [O(\mathfrak{T}_0^k, \boldsymbol{\mathfrak{T}^k}), \cdots, O(\mathfrak{T}_i^k, \boldsymbol{\mathfrak{T}^k}), \ldots, O(\mathfrak{T}_{mn-1}^k, \boldsymbol{\mathfrak{T}^k})]$. Using the estimated reordering sequence $\hat{\boldsymbol{R}}^{\boldsymbol{k}}$, we generate a new set of positional features, $\tilde{\boldsymbol{\mathcal{E}}}^{\boldsymbol{k}}$, by indexing the original positional feature vector set $\boldsymbol{\mathcal{E}^k}$ in the order defined by $\hat{\boldsymbol{R}}^{\boldsymbol{k}}$. More precisely, $\tilde{\boldsymbol{\mathcal{E}}}_{\boldsymbol{i}}^{\boldsymbol{k}} = \boldsymbol{\mathcal{E}}_{\hat{\boldsymbol{R}}_{\boldsymbol{i}}^{\boldsymbol{k}}}^{\boldsymbol{k}}$ for $i \in [0, mn)$. Then, the estimate of the permutation matrix $\boldsymbol{\mathcal{R}}^{-1}$, denoted $\hat{\boldsymbol{\mathcal{R}}}^{\boldsymbol{k}}$, can be computed by determining the cosine similarity between the elements in $\tilde{\boldsymbol{\mathcal{E}}}^{\boldsymbol{k}}$ and $\boldsymbol{\mathcal{E}^k}$. The estimate $\hat{\mathcal{R}}_{i,j}^k$ can be represented as follows:

$$\hat{\mathcal{R}}_{i,j}^k = \frac{\tilde{\boldsymbol{\mathcal{E}}}_{\boldsymbol{i}}^{\boldsymbol{k}} \cdot \boldsymbol{\mathcal{E}}_{\boldsymbol{j}}^{\boldsymbol{k}}}{\left\|\tilde{\boldsymbol{\mathcal{E}}}_{\boldsymbol{i}}^{\boldsymbol{k}}\right\| \cdot \left\|\boldsymbol{\mathcal{E}}_{\boldsymbol{j}}^{\boldsymbol{k}}\right\|}. \tag{7}$$

It is notable that all operations following the acquisition of the positional feature vectors $\boldsymbol{\mathcal{E}^k}$ do not involve unknown trainable parameters. Therefore, the argsort operation as performed here can be used to effectively hinder gradient backpropagation. The final estimate of the permutation matrix obviously satisfies the condition that the maximum value in each row is not in the same column. By performing a softmax operation for each row, the final prediction of the permutation matrix can be achieved. In the same way, estimates of the permutation matrix $\hat{\boldsymbol{\mathcal{R}}}^{\boldsymbol{h}}$ can be derived from feature maps at other levels.

C. Loss Function Design

The permutation matrix estimate derived in Section III-B, $\hat{\boldsymbol{\mathcal{R}}}$, ensures that the maximum values in each row are not aligned in the same column. If we consider only the position combinations that correspond to the maximum values, these combinations represent all possible permutations of the image segmentation strategy. This approach allows using the network to explore all possibilities of segmentation strategies and dynamically output permutation matrix predictions with different shapes based on the image segmentation strategy. In this section, the loss function of the network is defined from two perspectives: the relative position and the absolute position of the image blocks. The approaches based on these different perspectives allow the network to focus not only on the adjacency relationships between the blocks but also on the absolute position of the image blocks.

First, we can use the permutation matrix estimate $\hat{\boldsymbol{\mathcal{R}}}$ to derive the recovery image of the permuted image. By left-multiplying the original disordered block set $\tilde{\boldsymbol{P}}$ by $\hat{\boldsymbol{\mathcal{R}}}$, we can recover the positions of the image blocks and reassemble them into a new recovery image, $\hat{P}$. This process can be expressed by the following formula:

$$\begin{pmatrix} \hat{\mathcal{P}}_0 \\ \vdots \\ \hat{\mathcal{P}}_i \\ \vdots \\ \hat{\mathcal{P}}_{mn-1} \end{pmatrix} = \hat{\boldsymbol{\mathcal{R}}} \times \begin{pmatrix} \tilde{\mathcal{P}}_0 \\ \vdots \\ \tilde{\mathcal{P}}_i \\ \vdots \\ \tilde{\mathcal{P}}_{mn-1} \end{pmatrix}. \tag{8}$$

We define a set of adjacent blocks $\boldsymbol{U}(\hat{\mathcal{P}}_i)$, which is the set of image blocks adjacent to the image block $\hat{\mathcal{P}}_i$ in the recovery image $\hat{P}$. More precisely, $u(i)$, $d(i)$, $l(i)$, and $r(i)$, respectively, represent the indices of the top, bottom, left, and right neighboring blocks of the image block with index i. The set of neighboring blocks $\boldsymbol{U}(\hat{\mathcal{P}}_i)$ can be expressed by the following formula: $\boldsymbol{U}(\hat{\mathcal{P}}_i) = \{\hat{\mathcal{P}}_{u(i)}, \hat{\mathcal{P}}_{d(i)}, \hat{\mathcal{P}}_{l(i)}, \hat{\mathcal{P}}_{r(i)}\}$. The definitions for $u(i)$, $d(i)$, $l(i)$, and $r(i)$ are as follows:

$$\begin{aligned} u(i) &= \begin{cases} i-n, & \text{if } i \geq n \\ mn, & \text{otherwise} \end{cases} \\ d(i) &= \begin{cases} i+n, & \text{if } i < mn-n \\ mn, & \text{otherwise} \end{cases} \\ l(i) &= \begin{cases} i-1, & \text{if } i\%n > 0 \\ mn, & \text{otherwise} \end{cases} \\ r(i) &= \begin{cases} i+1, & \text{if } i\%n < n-1 \\ mn, & \text{otherwise.} \end{cases} \end{aligned} \tag{9}$$

In this set, $\hat{\mathcal{P}}_{u(i)}$, $\hat{\mathcal{P}}_{d(i)}$, $\hat{\mathcal{P}}_{l(i)}$, and $\hat{\mathcal{P}}_{r(i)}$ represent the adjacent image blocks of $\hat{\mathcal{P}}_i$ in the up, down, left, and right directions, respectively. For the image blocks at the edge of the recovery image $\hat{P}$, there may not be neighboring blocks in some directions. These nonexistent neighbor blocks are instead represented as an image block $\hat{\mathcal{P}}_{mn}$ that has the same shape as $\hat{\mathcal{P}}_i$, but all elements are 0. In this way, regardless of whether the image block $\hat{\mathcal{P}}_i$ is on the boundary, we can obtain its four neighbor blocks in different directions through the set $\boldsymbol{U}(\hat{\mathcal{P}}_i)$, which facilitates subsequent processing and analysis.

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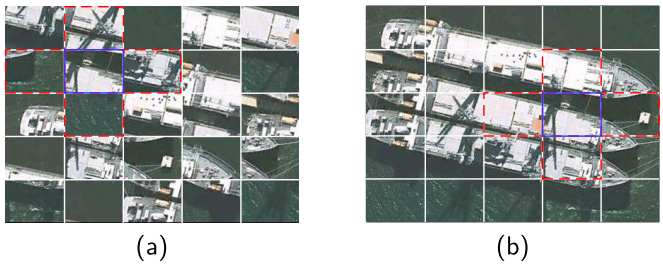


Fig. 5. Illustration of the adjacent regression loss function. (a) Predicted image with one image block highlighted by a blue solid line and its adjacent blocks marked by red dashed lines. (b) Original image with the image block is most similar to the predicted image outlined by a blue solid line and its adjacent blocks marked by red dashed lines. The proposed relative position loss function is defined as the $L2$ norm of the differences between these two sets of neighboring blocks.

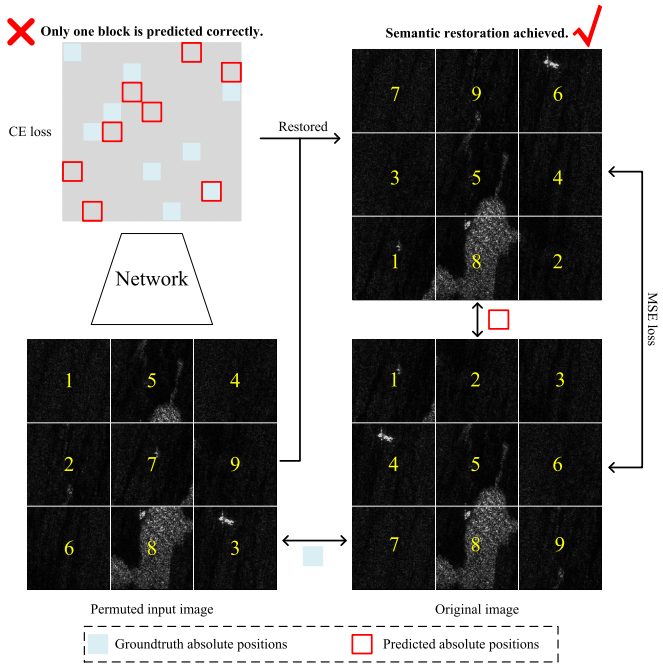


Fig. 6. Comparative illustration of two different methods for absolute positioning. Constraint of the predicted absolute position by a CE loss function is shown on the left, while application of an MSE loss function directly between the recovered and original images is shown on the right. The figure demonstrates that the first method semantically treats similar image blocks that are misplaced during prediction as prediction errors, while the second method treats them as nearly correct and dynamically focuses on image blocks with larger differences.

As shown in Fig. 5, for each predicted image block $\hat{\mathcal{P}}_i$, we can find the most similar image block $\mathcal{P}_j$ in the original image P by minimizing the mean squared error (MSE) between the predicted image block $\hat{\mathcal{P}}_i$ and the image block $\mathcal{P}_j$ of the original image. The same approach gives us the set of neighboring blocks $\boldsymbol{U}(\mathcal{P}_j) = \{\mathcal{P}_{u(j)}, \mathcal{P}_{d(j)}, \mathcal{P}_{l(j)}, \mathcal{P}_{r(j)}\}$ for block $\mathcal{P}_j$ in the original image P. Therefore, we define the adjacent regression CE loss function for block $\hat{\mathcal{P}}_i$ as follows:

$$L_r^{(i)} = \text{MSE}\big\{\boldsymbol{U}(\hat{\mathcal{P}}_i), \boldsymbol{U}(\mathcal{P}_j)\big\} \tag{10}$$

where $j = \underset{k}{\text{argmin}}\{\text{MSE}(\hat{\mathcal{P}}_i, \mathcal{P}_k)\}$. Moreover, we can also constrain the absolute position of the block. As can be seen in Fig. 6, we highlight the stark contrast between the two different methods of determining absolute position. Illustrated on the left side is the use of a CE loss function that acts directly on the predicted absolute position. In contrast, illustrated on the right side is the use of an MSE loss function applied directly between the recovered and original images. Specifically, the incorrect positioning of semantically similar image blocks, which is a common occurrence in remote sensing imagery, is interpreted as a prediction error in the first method. In contrast, the second method treats them as nearly correct, with the added benefit of dynamically focusing on image blocks that exhibit greater differences. For the permutation matrix shown in Fig. 6, the highlighted entry in the ith row and jth column means that the jth image block is positioned at the location of the ith block in the original image. To maximize the benefits of both methods in this work, we use a weighted approach, where the MSE method is weighted more heavily due to its dynamic focusing ability on image blocks with larger differences. The loss function for the absolute position of the block is defined as follows:

$$\begin{aligned} L_a^{(i)} = 0.2 \times \frac{1}{mn} \sum_{k=0}^{mn-1} \big\{ &(\mathcal{R}^{-1})_{i,k} \log \hat{\mathcal{R}}_{i,k} \\ &+ (1 - (\mathcal{R}^{-1})_{i,k}) \log(1 - \hat{\mathcal{R}}_{i,k}) \big\} \\ &+ \text{MSE}(\hat{\mathcal{P}}_i, \mathcal{P}_i). \end{aligned} \tag{11}$$

Since the constraints on image block recombination are overly strict, we randomly select only a portion β of the blocks for constraint. Therefore, the final loss function can be expressed as follows:

$$L = \frac{1}{\lceil mn\beta \rceil} \sum_{i \in \boldsymbol{I}} (L_r^{(i)} + L_a^{(i)}) \tag{12}$$

where $\boldsymbol{I}$ is a subset of the set $\{0, 1, \ldots, mn-1\}$ satisfying the condition $|\boldsymbol{I}| = \lceil mn\beta \rceil$ in which $|\boldsymbol{I}|$ denotes the number of elements in the set $\boldsymbol{I}$ and $\lceil \cdot \rceil$ denotes rounding up to the nearest integer. In our experiments, the parameter β is set to 0.25.

IV. Experiments

To thoroughly test the effectiveness of the proposed model, we set out on the challenging task of oriented object detection. In contrast to the detection of horizontal objects, the detection of rotated objects poses a significant challenge. Unlike horizontal objects, rotated objects can appear in any orientation, which dramatically increases the complexity of the detection task. This complexity stems from the need to detect and correctly interpret objects of different shapes, sizes, and orientations, which can significantly affect detection accuracy.

Given the inherent difficulties in detecting rotated objects, we strategically chose to conduct our experiments on two different remote sensing image datasets—HRSC2016, an optical remote sensing image dataset, and RSDD, an SAR image dataset. Both datasets are characterized by their focus on oriented object detection, which makes them ideal platforms for testing our proposed model. Our experimental results show that the detection performance of most existing oriented object detection methods can be effectively improved by using the scheme proposed in this study. These improvements demonstrate that our approach not only improves the interpretability of remote sensing images but also provides a robust and reliable solution for object recognition in various imaging contexts.

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A. Datasets

The HRSC2016 dataset, a renowned and comprehensive resource, is widely used for complex tasks such as arbitrary orientation vessel detection. It consists of 1061 images of different sizes, ranging from 300×300 to 1500×900 pixels, and provides a wide range of different ship types and orientations. The dataset is methodically split into a training set of 436 images and a validation set of 181 images, both of which are used for training the model, while the remaining images serve as a test set. Using the HRSC2016 dataset for validation offers several advantages. First, the considerable variation in image sizes provides a unique opportunity to evaluate our methodology in terms of its scalability and adaptability to different resolutions, a critical property for any robust object recognition model. Second, the focus on detecting vessels of arbitrary orientation within the dataset challenges the model to accurately identify and classify vessels in a wide range of orientations, ultimately increasing its versatility and robustness. We chose mean average accuracy (mAP) as the measure for evaluating the detection accuracy based on the HRSC2016 dataset. This measure is consistent with established evaluation standards as defined by PASCAL VOC 2007 [30] and VOC 2012. This standardized and consistent evaluation measure allows for a reliable comparison of the performance of our model with those of previous methods and provides an objective assessment of its effectiveness and competitiveness.

The SODA-A dataset, with its extensive assembly of high-resolution aerial imagery, includes a vast range of urban landscapes, amassing 872 069 instances over nine distinct classes. The dataset's strength lies in its diverse representation of urban density, with the instances per image varying dramatically from a solitary object to an overwhelming 11 134, averaging 318. This diversity ensures the dataset's applicability across varying scenarios, from sparse to dense object congregations, which poses unique challenges in object detection. For a detailed assessment of object detection capabilities, nuanced area-based stratification is introduced in SODA-A for calculating average precision (AP)—AP_{eS} (extremely small), AP_{rS} (relatively small), AP_{gS} (generally small), and AP_N (normal)—covering a range from less than 144 pixels to 2000 pixels in the object area. This categorization facilitates the refined analysis of detection algorithms, assessing their precision and generalizability across different object sizes, which are crucial aspects of aerial image interpretation. The primary metric, overall AP, is derived by averaging APs across ten intersection over union (IoU) thresholds from 0.5 to 0.95, with a specific focus on small objects. AP_{50} and AP_{75} are determined at IoU thresholds of 0.5 and 0.75, respectively, providing detailed insights into the detection accuracy. Additionally, the AP metrics for the aforementioned area subsets underscore our focus on accurately detecting objects of limited size, further demonstrating the robustness of our evaluation approach.

The RSDD dataset is a comprehensive compilation of 127 scenes carefully gathered from a variety of high-resolution data sources. These include 84 scenes derived from the GF-3 satellite, 41 scenes from TerraSAR-X data, and two untrimmed large-scale images. The dataset includes a variety of imaging modes, polarization methods, and resolution variations, culminating in a comprehensive array of 7000 slices with more than 10 263 ship representations. The labels of this dataset were carefully created using a balanced combination of automatic labeling and manual corrections, enhancing the precision and credibility of the dataset. The RSDD dataset is distinguished by a number of unique features, including different directions of rotation, significant aspect ratios, a high prevalence of smaller targets, and a selection of different scenes. These eclectic elements provide a comprehensive simulation of real-world conditions, which strengthens the resilience of any model trained using this dataset. The dataset consists of 7000 images divided into a training set of 5000 images and a test set of 2000 images. Furthermore, the test set is divided into two different scenarios, inshore and offshore, with 159 and 1841 images, respectively. This deliberate division provides a heterogeneous test environment that reflects different real-world scenarios and allows a comprehensive evaluation of the model's performance and adaptability.

B. Parameter Settings

Our experiments were conducted on a robust hardware platform hosted on a server running the CentOS Linux operating system. This server is powered by an AMD EPYC 7763 64-core processor with a CPU frequency of 2.45 GHz. To meet the demanding processing requirements of our work, we used an NVIDIA A100-SXM4-80GB graphics card. To ensure a consistent and reproducible environment for our work, we used docker technology to create a virtual runtime environment. This approach provides an isolated sandbox that mitigates issues related to system variations and dependencies and greatly improves the reproducibility of our experimental results. This docker environment was based on the dev-1.x branch of the mmrotate [31] platform. In this environment, we used Python 3.7.11 in conjunction with CUDA 11.3 for GPU-accelerated computations. Image preprocessing and enhancement were performed using PyTorch version 1.13.1 + cu117, augmented by TorchVision 0.11.0. OpenCV 4.7.0 was integrated to handle a wide range of image processing and computer vision tasks. To enable distributed computing, we used the NCCL backend in our docker configuration. The multiprocess settings were adjusted to select "fork" as the startup method and set the number of threads to 0 for OpenCV. To ensure consistent experimental results, we disabled the CuDNN performance benchmark. To ensure the stability and reproducibility of our experiments in this docker-based environment, we set the random value to 0. It is worth noting that we did not assign a different value for each rank throughout the experiment. This docker-based experimental setup, together with the described system configuration, provided a stable and efficient platform for our research.

To ensure comprehensive coverage, we used the ResNet series [32] and feature pyramid network (FPN) [33] as our primary networks for feature extraction. First, these networks were pretrained on ImageNet to provide a solid foundation for our study. Based on this, we further fine-tuned these networks in the corresponding datasets using the proposed PIR self-supervised method. In the described configuration, we use a 3×3 segmentation strategy for image partitioning

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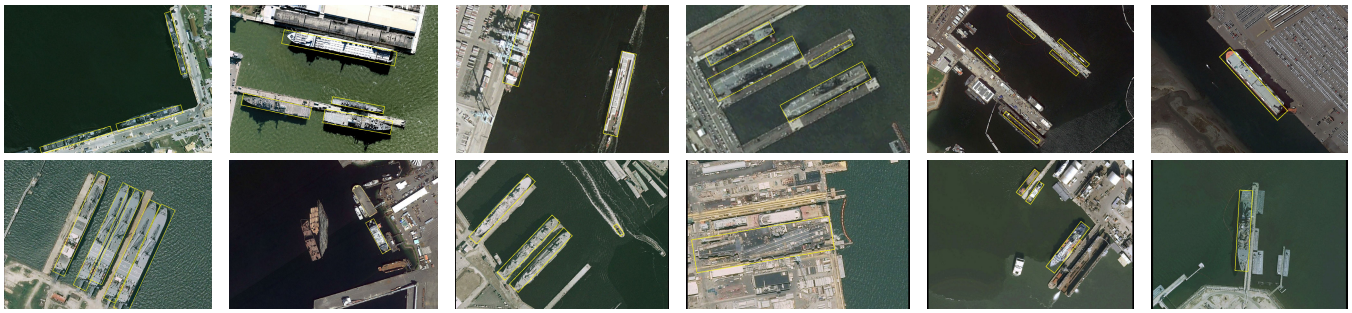


Fig. 7. Examples of detection results on the HRSC2016 dataset using oriented R-CNN with an R-50-FPN backbone trained using the proposed PIR self-supervised method. The oriented bounding boxes with scores above 0.3 are shown. Black padding is added to some images for a uniform layout. For enhanced clarity and details, consider enlarging the image.

by default. For the two parameters of the position decoding methods and the decoding dimensions in the permutation matrix estimation module, we used our proposed IPD approach with a dimensionality of 16. The default training procedure spans 50 epochs. With these configurations, we further refined these networks using the proposed self-supervised PIR method on the corresponding datasets. This rigorous training process was accelerated by the use of eight NVIDIA A100-SXM4-80 GB graphics cards, each of which simultaneously processed a batch size of 128, significantly increasing the efficiency of our computational workflow. Self-supervised training was based on the use of multiscale training and random flipping as data augmentation techniques. Our network was optimized using the stochastic gradient descent (SGD) algorithm with a learning rate of 0.02, a momentum of 0.9, and a weight descent of $5e^{-5}$. The learning strategy was divided into two phases. In the first phase, we used a linear learning rate (LinearLR) scheduler with a starting factor of 0.1 per epoch, which included the first five epochs. Then, a cosine annealing learning rate (CosineAnnealingLR) scheduler was applied, characterized by a maximum period value of 45. This scheduler was also applied to each epoch and spanned the subsequent epochs from the 5th to the 50th epoch. This two-stage learning strategy allowed for a dynamic and adjustable learning rate that was calibrated to match the progression of the training process and improve the model performance. After individually training the feature extraction networks using our PIR method, the refined weights were applied to the downstream task of oriented object detection. This application was performed on two datasets with different parameter configurations.

For the HRSC2016 and SODA-A datasets, we applied horizontal and vertical flipping as data augmentation techniques during training. The entire network was optimized using the SGD algorithm with a momentum of 0.9 and a weight descent of 0.0001. At this stage, we maintained the aspect ratios of the images. Throughout 36 epochs, we implemented an initial learning rate of 0.005, which was then divided by 10 at the 24th and 33rd epochs.

We then used random horizontal flipping with a probability of 0.5 as a data augmentation technique for the RSDD dataset. Our network was optimized using the SGD algorithm with an initial learning rate of 0.0025, a momentum of 0.9, and a weight decay of 0.0001. $L2$ norm gradient pruning was applied, with the maximum gradient norm set to 35. This process was iterated over 12 epochs, with a tenfold reduction in the learning rate at the 7th and 10th epochs. During the first 500 batches, a linear warm-up learning rate was applied, set to one-third of the initial learning rate. The batch size was adjusted depending on the backbone network used: When ResNet-18 was used, a batch size of 8 was used, while for ResNet-34, ResNet-50, ResNet-101, and ResNet-152, the batch size was set to 4. All training and test images were uniformly set to a resolution of 512×512 pixels, and the confidence threshold for the test was set to 0.05.

C. Experiments With HRSC2016 and SODA-A

In our research, various oriented object detection methodologies were rigorously tested on the HRSC2016 and SODA-A optical remote sensing image datasets. Within the HRSC2016 dataset, we gauged the efficacy of these methods through mean AP (mAP), adhering to the established PASCAL VOC 2007 and 2012 criteria, as detailed in Table I. The findings reveal that the performance of oriented object detection networks is significantly bolstered when using our PIR self-supervised training approach. This enhancement is observed across different detection methods and feature extraction network scales, despite our network being pretrained for only 50 epochs with a fundamental 3×3 block strategy, and without reliance on human annotation. The method's success lies in its exploitation of the intrinsic contextual relationships within the images, as evidenced by the precise localization of objects in various orientations and scales, as demonstrated in Fig. 7.

For the SODA-A dataset, as noted in Table II, we implemented a nuanced 7×7 block strategy to address the challenges presented by the dataset's smaller object sizes. This tailored approach facilitates enhanced detection of small-scale objects, as is necessary given the intricate nature of this dataset. We present the overall AP, AP_{50}, and AP_{75}, along with APs for different object sizes. Although the pretrained weights adapted via our method resulted in improvements in detection performance, the enhancements were marginal. Given the ample size of the SODA-A dataset, the conventional detection methods are already well-equipped to mine the subtle features of small objects in remote sensing images. Additionally, the predominantly small scale of targets in SODA-A implies that global positional information, learned through our method, is not pivotal for these minuscule targets that are heavily reliant on local cues. Consequently, while our method adds value, it serves more as a refinement, rather than a transformational change for the SODA-A dataset.

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TABLE I
EXPERIMENTAL RESULTS OF VARIOUS MODELS ON THE TEST SUBSET OF THE HRSC2016 DATASET

Method	Publication	Backbone	mAP(07)		mAP(12)	
			Baseline	Proposed PIR	Baseline	Proposed PIR
RoI Transformer [25]	CVPR 2019	R-101	86.20	86.72(↑ **0.52**)	-	-
Gliding Vertex [34]	TPAMI 2021	R-101	88.20	88.65(↑ **0.45**)	-	-
CenterMap Net [35]	TGRS 2021	R-50	-	-	92.80	93.06(↑ **0.26**)
S2ANet [23]	TGRS 2022	R-101	90.17	90.79(↑ **0.62**)	95.01	95.37(↑ **0.36**)
DAL [36]	AAAI 2021	R-101	89.77	90.29(↑ **0.52**)	-	-
Oriented RCNN [24]	ICCV 2021	R-101	90.50	90.92(↑ **0.42**)	97.60	97.94(↑ **0.34**)
R3Det [37]	AAAI 2021	R-101	89.26	89.68(↑ **0.42**)	96.01	96.24(↑ **0.23**)
DODet [38]	TGRS 2022	R-101	90.18	90.57(↑ **0.39**)	95.84	96.10(↑ **0.26**)
AOPG [39]	TGRS 2022	R-50	90.34	90.71(↑ **0.37**)	96.22	96.39(↑ **0.17**)
QPDet [40]	TGRS 2023	R-50	90.47	90.90(↑ **0.43**)	96.60	96.81(↑ **0.21**)

† indicates that the reported result is directly copied from the referenced paper.

TABLE II
EXPERIMENTAL RESULTS OF VARIOUS MODELS ON THE TEST SUBSET OF THE SODA-A DATASET

Pretrained	Method	Publication	AP	AP_{50}	AP_{75}	AP_{eS}	AP_{rS}	AP_{gS}	AP_N	#Param.	FLOPS
Baseline	RoI Transformer [25]	CVPR 2019	36.0	73.0	30.1	13.5	30.3	46.1	39.5	55.08M	306.20G
	Gliding Vertex [34]	TPAMI 2021	31.7	70.8	22.6	11.7	27.0	41.1	33.8	41.14M	292.25G
	R-Faster-RCNN [41]	TPAMI 2017	32.5	70.1	24.3	11.9	27.3	42.2	34.4	41.14M	292.25G
	S2ANet [23]	TGRS 2022	28.3	69.6	13.1	10.2	22.8	35.8	29.5	38.64M	732.74G
	R-RetinaNet [42]	TPAMI 2020	26.8	63.4	16.2	9.1	22.0	35.4	28.2	36.16M	800.21G
	Oriented RCNN [24]	ICCV 2021	34.4	70.7	28.6	12.5	28.6	44.5	36.7	41.13M	292.44G
	DHRec [43]	TPAMI 2022	30.1	68.8	19.8	10.6	24.6	40.3	34.6	31.99M	792.76G
	DODet [38]	TGRS 2022	31.6	68.1	23.4	11.3	26.3	41.0	33.5	69.34M	555.49G
	CFINet [44]	ICCV 2023	34.4	73.1	26.1	13.5	29.3	44.0	35.9	43.98M	305.94G
Proposed PIR	RoI Transformer [25]	CVPR 2019	36.5 (↑ **0.5**)	74.3 (↑ **1.3**)	30.8 (↑ **0.7**)	13.9 (↑ **0.4**)	31.2 (↑ **0.9**)	47.2 (↑ **1.1**)	40.0 (↑ **0.5**)	55.08M	306.20G
	Gliding Vertex [34]	TPAMI 2021	32.0 (↑ **0.3**)	71.1 (↑ **0.3**)	23.6 (↑ **1.0**)	11.8 (↑ **0.1**)	27.8 (↑ **0.8**)	41.8 (↑ **0.7**)	33.9 (↑ **0.1**)	41.14M	292.25G
	R-Faster-RCNN [41]	TPAMI 2017	32.9 (↑ **0.4**)	71.6 (↑ **1.5**)	24.5 (↑ **0.2**)	12.1 (↑ **0.2**)	27.6 (↑ **0.3**)	42.9 (↑ **0.7**)	34.7 (↑ **0.3**)	41.14M	292.25G
	S2ANet [23]	TGRS 2022	29.2 (↑ **0.9**)	70.8 (↑ **1.2**)	14.4 (↑ **1.3**)	10.7 (↑ **0.5**)	23.2 (↑ **0.4**)	36.5 (↑ **0.7**)	30.4 (↑ **0.9**)	38.64M	732.74G
	R-RetinaNet [42]	TPAMI 2020	27.7 (↑ **0.9**)	64.7 (↑ **1.3**)	17.2 (↑ **1.0**)	9.3 (↑ **0.2**)	22.6 (↑ **0.6**)	36.0 (↑ **0.6**)	29.0 (↑ **0.8**)	36.16M	800.21G
	Oriented-RCNN [24]	ICCV 2021	35.2 (↑ **0.8**)	71.6 (↑ **0.9**)	30.0 (↑ **1.4**)	13.0 (↑ **0.5**)	29.3 (↑ **0.7**)	44.7 (↑ **0.2**)	37.7 (↑ **1.0**)	41.13M	292.44G
	DHRec [43]	TPAMI 2022	30.7 (↑ **0.6**)	69.6 (↑ **0.8**)	20.7 (↑ **0.9**)	11.1 (↑ **0.5**)	25.4 (↑ **0.8**)	41.3 (↑ **1.0**)	35.1 (↑ **0.5**)	31.99M	792.76G
	DODet [38]	TGRS 2022	31.9 (↑ **0.3**)	68.9 (↑ **0.8**)	23.9 (↑ **0.5**)	11.5 (↑ **0.2**)	27.0 (↑ **0.7**)	41.6 (↑ **0.6**)	33.7 (↑ **0.2**)	69.34M	555.49G
	CFINet [44]	ICCV 2023	34.9 (↑ **0.5**)	74.0 (↑ **0.9**)	26.8 (↑ **0.7**)	14.3 (↑ **0.8**)	29.7 (↑ **0.4**)	44.7 (↑ **0.7**)	36.5 (↑ **0.6**)	43.98M	305.94G

D. Experiment With RSDD

Compared with optical remote sensing imagery, target detection in SAR images can be particularly challenging, especially in complex background scenarios. In addition, the need to detect oriented targets further increases the difficulty of the task due to the stringent requirements for proper target orientation. The ships in the RSDD dataset represent different rotational directions, high aspect ratios, a high proportion of small targets, and a wide variety of scenes. The ability to effectively improve target detection performance in such a challenging dataset is key to demonstrating the effectiveness of the proposed model. Therefore, we compare the detection performance of different oriented object detection networks with the ResNet50 FPN feature extraction network, both before and after applying our proposed PIR method, to verify its effectiveness. Specifically, we use various popular oriented object detection models, including single-stage and two-stage networks, to evaluate the AP at different AP thresholds. The detailed recognition performance results are shown in Table III. After pretraining with our proposed PIR self-supervised method, we observed significant improvements in our detection performance based on rigorous metrics, such as AP80, which means that our feature extraction network has become more accurate in target localization when using the PIR self-supervised method. In addition, due to the complexity of near-shore scenarios, all baseline methods generally performed moderately. After pretraining with the proposed PIR self-supervised method, the feature extraction network exhibited a significantly improved understanding of the distribution of vessels along the coast, resulting in significant performance improvements in inshore scenarios compared with offshore scenarios. The detection results for different backgrounds are shown in Fig. 8. Correctly detected targets are indicated by yellow-oriented rectangles, missed targets by red-oriented rectangles, and false positives by red-oriented ellipses. The visualized results impressively demonstrate the effectiveness of the proposed method.

In addition, we performed oriented object detection on an uncropped Gaofen-3 satellite image from the RSDD dataset. The image used was acquired on June 19, 2021, and is from the Gaofen-3 sensor positioned at E119.8 and N35.2. The imaging parameters were set to the FSI imaging mode with DV polarization and a product level of L1A. The image with an incidence angle of 19°–50° has a width of 50 km and a resolution of 5 m, corresponding to a size of $14\,848 \times 40\,448$ pixels. Table IV summarizes the test results on the uncropped GF3 image in the RSDD dataset. This image contains 176 labeled targets, of which our method correctly identified 153 and incorrectly identified 19. Targets were missed mostly due to their small size. Since most of the ship targets in the image are represented by elongated oblique rectangles, which are more difficult to localize than horizontal boxes, an accuracy of 89% under such high-level requirements demonstrates that the intrinsic image properties were learned from the feature extraction network using the proposed PIR method. This is a clear indication of the robustness and generalizability of our proposed model. The visualization of the recognition results in this image can be found in Fig. 9. Correctly detected targets are indicated by yellow-oriented rectangles, missed targets by red-oriented rectangles, and false positives by red-oriented ellipses.

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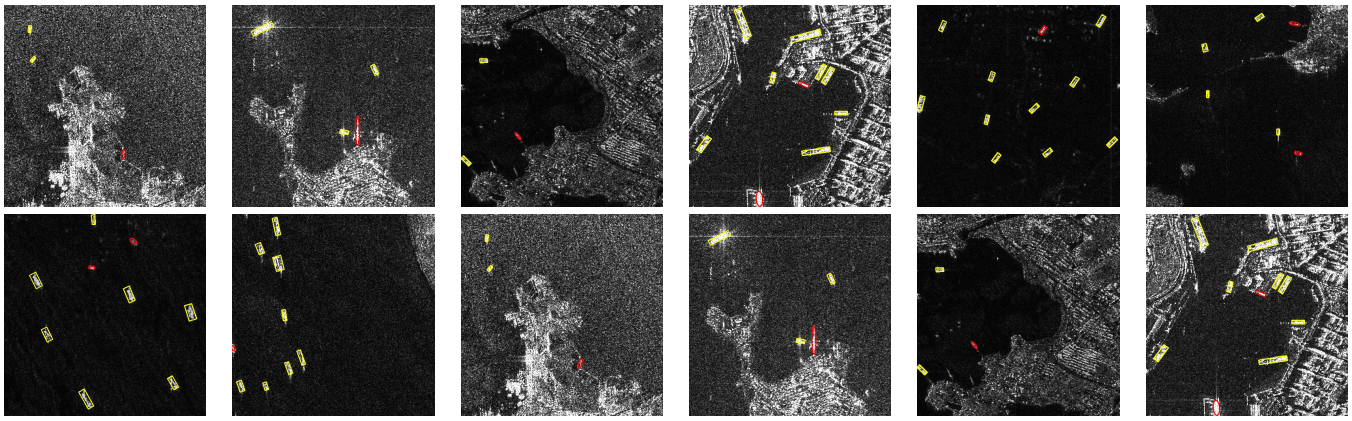


Fig. 8. Examples of detection results on the RSDD dataset using oriented R-CNN with an R-50-FPN backbone trained using the proposed PIR self-supervised method. The oriented bounding boxes with scores above 0.3 are shown. Correctly detected targets are indicated by yellow-oriented rectangles, missed targets by red-oriented rectangles, and false positives by red-oriented ellipses. For enhanced clarity and details, consider enlarging the image.

TABLE III

EXPERIMENTAL RESULTS OF VARIOUS MODELS ON THE TEST SUBSET OF THE RSDD DATASET

Pretrained	Method	Publication	AP_{50}(%)	Inshore			Offshore		
				AP_{50}(%)	AP_{65}(%)	AP_{80}(%)	AP_{50}(%)	AP_{65}(%)	AP_{80}(%)
Baseline	R-Faster-RCNN [41]	TPAMI 2017	78.9	50.8	30.1	8.1	88.7	65.4	12.3
	RoI Transformer [25]	CVPR 2019	86.8	54.9	40.2	8.5	89.8	76.8	18
	R-Reppoints [45]	ICCV 2019	76.1	43.3	26.3	5.8	86.9	60.9	17.3
	R-RetinaNet [42]	TPAMI 2020	78.6	53.1	37.8	3.9	88.8	72.9	16.2
	Oriented RCNN [24]	ICCV 2021	87.8	62.6	49.3	14.8	80.8	77.3	31.3
	CFA [26]	CVPR 2021	87.3	60.6	42.9	5.1	89.6	74.3	25.5
	S2ANet [23]	TGRS 2022	59.8	34.3	12.2	1.8	69.3	28.1	2
	DODet [38]	TGRS 2022	83.7	53.6	39.1	7.9	87.6	70.4	26.6
	AOPG [39]	TGRS 2022	86.4	54.7	39.4	8.3	89.2	73.1	29.7
	QPDet [40]	TGRS 2023	82.1	53.3	38.6	8.1	90.3	73.9	30.4
Proposed PIR	R-Faster-RCNN [41]	TPAMI 2017	79.6(↑ **0.7**)	52.0(↑ **1.2**)	32.4(↑ **2.3**)	11.5(↑ **3.4**)	89.2(↑ **0.5**)	67.3(↑ **1.9**)	14.6(↑ **2.3**)
	RoI Transformer [25]	CVPR 2019	87.6(↑ **0.8**)	56.3(↑ **1.4**)	43.3(↑ **3.1**)	12.4(↑ **3.9**)	90.2(↑ **0.4**)	78.4(↑ **1.6**)	20.4(↑ **2.4**)
	R-Reppoints [45]	ICCV 2019	77.0(↑ **0.9**)	44.7(↑ **1.4**)	28.0(↑ **1.7**)	8.0(↑ **2.2**)	87.4(↑ **0.5**)	62.9(↑ **2.0**)	20.1(↑ **2.8**)
	R-RetinaNet [42]	TPAMI 2020	79.4(↑ **0.8**)	54.3(↑ **1.2**)	39.4(↑ **1.6**)	6.4(↑ **2.5**)	89.1(↑ **0.3**)	74.2(↑ **1.3**)	18.3(↑ **2.1**)
	Oriented RCNN [24]	ICCV 2021	88.5(↑ **0.7**)	63.9(↑ **1.3**)	51.6(↑ **2.3**)	17.9(↑ **3.1**)	90.2(↑ **9.4**)	78.9(↑ **1.6**)	33.7(↑ **2.4**)
	CFA [26]	CVPR 2021	87.9(↑ **0.6**)	61.8(↑ **1.2**)	45.5(↑ **2.6**)	8.3(↑ **3.2**)	89.8(↑ **0.2**)	75.7(↑ **1.4**)	28.4(↑ **2.9**)
	S2ANet [23]	TGRS 2022	60.7(↑ **0.9**)	35.9(↑ **1.6**)	14.6(↑ **2.4**)	4.8(↑ **3.0**)	69.6(↑ **0.3**)	29.7(↑ **1.6**)	3.9(↑ **1.9**)
	DODet [38]	TGRS 2022	84.5(↑ **0.8**)	55.1(↑ **1.5**)	41.0(↑ **1.9**)	10.8(↑ **2.9**)	88.0(↑ **0.4**)	72.1(↑ **1.7**)	29.2(↑ **2.6**)
	AOPG [39]	TGRS 2022	87.4(↑ **1.0**)	56.4(↑ **1.7**)	41.5(↑ **2.1**)	11.8(↑ **3.5**)	89.6(↑ **0.4**)	74.5(↑ **1.4**)	32.4(↑ **2.7**)
	QPDet [40]	TGRS 2023	83.0(↑ **0.9**)	54.6(↑ **1.3**)	40.4(↑ **1.8**)	11.3(↑ **3.2**)	90.8(↑ **0.5**)	75.4(↑ **1.5**)	32.7(↑ **2.3**)

TABLE IV

TEST RESULTS ON THE UNCROPPED GF3 IMAGE IN THE RSDD DATASET

Detected Ships	True Ships	False Ships	Missed Ships	Precision	Recall
172	153	19	23	89.0%	86.9%

Given the size of the image, we selectively enlarged eight subregions for clarity. For a clearer view of the recognition results, we suggest zooming in and rotating the image as needed.

E. Ablation Experiment

To thoroughly investigate the potential of our proposed method, we chose the best detection model from our comparison experiments, the oriented R-CNN network-based object detection model. To ensure the comparison experiment was more representative, we chose the AP_{50} of the RSDD inshore test set as the comparison benchmark. We conducted extensive comparison experiments to appropriately analyze the application potential and robustness of the proposed model and determine its optimal configuration.

First, we analyzed the effects of the position decoding methods and the decoding dimensions in the permutation matrix estimation module on the extraction ability of the feature extraction network. We set the number of training iterations to 50, chose a 6×6 image partitioning strategy, and used ResNet-50 and FPN as feature extraction networks. The results of the experiment are shown in Table V. It can be seen that the output dimension of the perceptual module, which determines the vector of positional features, is a crucial parameter of the proposed network for estimating the permutation matrix. Ideally, this network aims to approximate the permutation matrix. Under perfect conditions, the vectors of positional features corresponding to the different image blocks should be orthogonal to each other. For a 6×6 partitioning strategy, the space of position vectors should have at least 36 dimensions to allow fully distinguishing different feature blocks, as lower values would limit the expressiveness of the network. As can be seen, the recognition performance of the perceptual modules of different position decoding methods drops drastically when the output dimensions are less than 36. When the output dimensions exceed 36, the recognition performance improves significantly due to the higher expressive power of the network. However, when the output dimensions are 128, the total prediction space becomes relatively sparse, which means that there is a higher training cost associated

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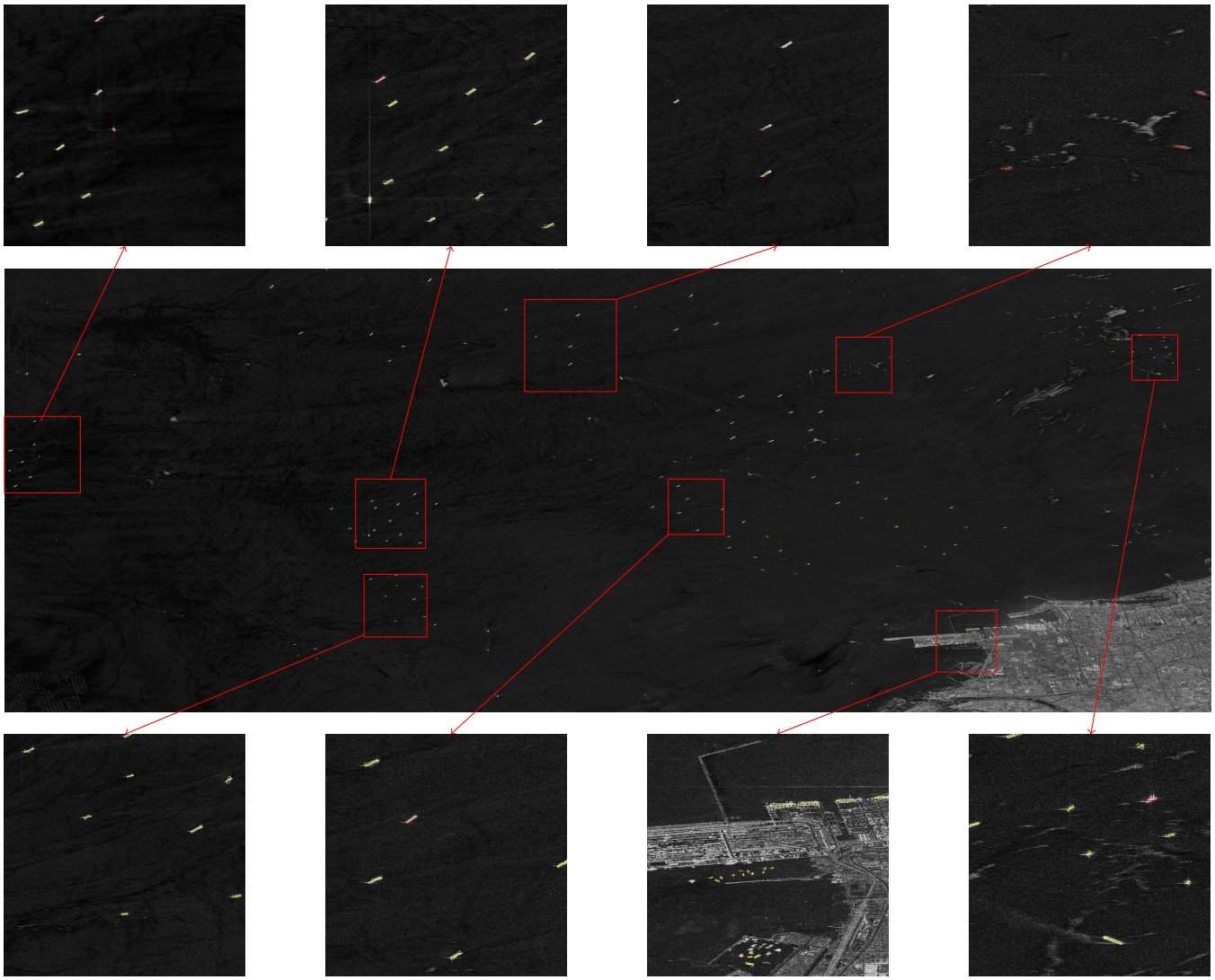


Fig. 9. Examples of detection results on the uncropped GF3 image in the RSDD dataset using oriented R-CNN with an R-50-FPN backbone trained using the proposed PIR self-supervised method. The oriented bounding boxes with scores above 0.3 are shown. Correctly detected targets are indicated by yellow-oriented rectangles, missed targets by red-oriented rectangles, and false positives by red-oriented ellipses. For enhanced clarity and details, consider enlarging the image.

TABLE V

EFFECTS OF DECODING METHODS AND DIMENSIONS ON $AP_{50}(\%)$ PERFORMANCE FOR INSHORE SHIP DETECTION

Decoding Method	16	32	64	128
Average	57.9	59.5	63.2	62.1
Maximum	57.2	59.1	62.8	61.6
Minimum	56.7	58.8	62.9	61.4
Proposed IPD	58.3	60.7	**65.9**	64.2

TABLE VI

COMPARATIVE ANALYSIS OF $AP_{50}(\%)$ PERFORMANCE FOR INSHORE SHIP DETECTION UNDER DIFFERENT IMAGE SEGMENTATION STRATEGIES AND NETWORK SCALES

Segmentation Strategy	R-18	R-34	R-50	R-101	R-152
3×3	62.1	62.4	63.9	64.4	64.5
4×4	61.8	62.7	64.1	64.8	65.3
5×5	60.3	62.5	64.5	65.2	65.9
Dynamic †	61.2	62.9	65.2	65.9	**66.2**

†: Dynamic means that a 3×3 segmentation strategy is used for the first 30 epochs, followed by a 5×5 segmentation strategy for the next 20 epochs.

with converging to the optimum, resulting in slightly lower performance compared with dimensions of 64. At the same time, the proposed IPD method we chose is inherently better at finding representative position values than simply finding the maximum, minimum, or average value. It is more informative and achieves the best detection performance of 65.9%.

The effects of image segmentation strategy and network scale on AP_{50} performance for inshore ship detection are examined in this analysis, as shown in Table VI. With the evolution from ResNet-18 to ResNet-152, an overall improvement in the AP_{50} value is observed, assuming the segmentation strategy remains the same. This demonstrates the ability of larger network architectures to capture more complex feature representations, thereby improving the performance in challenging pretext tasks. An interesting observation is the less significant improvement in performance as we move from ResNet-101 to ResNet-152. This suggests that after a certain network size is reached, the benefits of further scaling diminish. This could be due to the inadequacy of the training iterations (50 epochs), where larger networks may

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TABLE VII

COMPARATIVE ANALYSIS OF $AP_{50}(\%)$ PERFORMANCE FOR INSHORE SHIP DETECTION UNDER DIFFERENT LOSS FUNCTION COMPONENTS AND PORTION β OF BLOCKS FOR CONSTRAINT

L_r †	L_a ‡		Portion β of Blocks for Constraint		
	CE	MSE	0.3	0.6	0.9
✓			63.5	62.5	61.2
	✓		61.4	61.7	62.3
		✓	62.1	62.6	62.7
✓	✓	✓	**64.2**	62.3	60.9

†: L_r means relative positional loss function.
‡: L_a means absolute positional loss function.

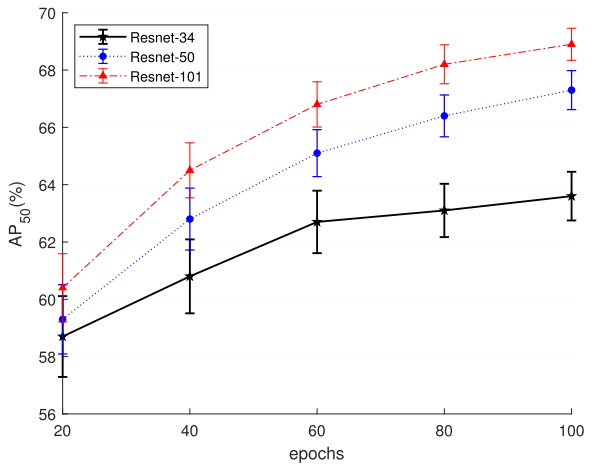


Fig. 10. Comparison of AP_{50} detection performance for inshore ship detection for ResNet-34, ResNet-50, and ResNet-101 under a 5×5 segmentation strategy over a different number of training epochs. It is evident that the detection performance improves as the number of training epochs increases, with the rate of performance increase slowing down for smaller models as they approach their expression capacity.

not use sufficient training time to fully learn and converge. When evaluating the impact of segmentation strategies, larger 5×5 segmentation strategies tend to improve performance more significantly than their smaller 3×3 counterparts. This implies that increasing the complexity of the pretext task using larger segmentation strategies can improve performance in cases of learning difficulty and, ultimately, the performance. Remarkably, our proposed dynamic segmentation strategy, which starts with a less complex 3×3 strategy and gradually increases to a 5×5 strategy during training, outperforms all other approaches. This approach maintains a manageable learning difficulty in the early stages of training and gradually increases the challenge, allowing the network to effectively adapt and learn throughout the training process.

Table VII delineates the impact of loss function components on the performance of object detection within inshore SAR images. The individual and combined effects of the three loss function components posited in this work are meticulously examined across varying proportions of constraint blocks. Analysis of the tabulated results reveals there is performance degradation when an excessive number of blocks are constrained using the relative positional loss function. This decline can be attributed to the intrinsic consideration of neighboring blocks in the loss function, where an overemphasis on local vicinity leads to counterproductive internal competition within the loss function mechanism. Notwithstanding, it is the relative positional loss component that most substantially elevates performance among the three that are considered. In contrast, the absolute positional loss, particularly when based on MSE, outperforms that based on CE loss. Furthermore, the absolute positional loss does not exhibit internal antagonism, indicating that increasing the number of constraint blocks tends to beneficially augment detection performance.

An experimental study was conducted to assess the influence of training duration on AP_{50} performance in coastal vessel detection, utilizing a 5×5 segmentation approach paired with varying ResNet architectures (34, 50, and 101 layers). To further evaluate the effect of random seeds on model performance, we standardized all other hyperparameters and executed 20 iterations with distinct seeds for each architecture, as depicted in Fig. 10. The mean and standard deviation of the performance metrics were computed, with the latter represented as error bands in Fig. 10. The findings reveal a pronounced correlation between training epochs and detection efficacy. Initially, at 20 epochs, performance across all architectures was suboptimal, occasionally even regressing below the baseline, likely due to the initial adjustment phase of the networks and the insufficient capture of image content nuances. However, with increasing epochs, all architectures exhibited a rapid enhancement in recognition ability. Notably, for ResNet-34, the performance plateaued after 60 epochs, suggesting that its capacity for representation may be limited under a 5×5 segmentation regime, as it reaches the point of learning saturation. In contrast, ResNet-50 and ResNet-101 demonstrated consistent advancements in performance with the extension of training epochs, underscoring the added benefit of the greater representational capabilities of larger models, which facilitate learning more intricate data features via complex pretext tasks. Additionally, error band analysis indicated the growing stability of model performance with extended training and larger network architectures, emphasizing the necessity of tailoring the training epoch count to the chosen architecture and segmentation strategy for optimal vessel recognition performance.

V. CONCLUSION

In this article, we propose a groundbreaking method integrating image permutation preprocessing, permutation matrix estimation, and bespoke loss function design. This method leverages PIR to augment the capabilities of feature extraction networks. The initial stage involves segmenting and shuffling input images into rearranged compositions, thus tailoring the difficulty of the pretext task to the evolving learning status of the networks. After preprocessing, the positional features extracted from feature maps inform permutation matrix estimation, a pivotal step in discerning relationships among feature blocks. Our innovative loss function, focusing on the relative and absolute positioning of blocks, judiciously imposes constraints on a randomly chosen subset, thus mitigating overconstraints and enhancing network convergence. The efficacy of our approach is validated by the empirical results obtained across various datasets.

Notwithstanding these advancements, limitations of the method were encountered. Its performance waned with small-sized or disproportionately represented targets,

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signifying areas ripe for improvement. Furthermore, incrementing the complexity of both the feature extraction network and image segmentation strategy initially escalated performance but eventually led to a plateau, highlighting the necessity to achieve a judicious balance between network intricacy and pretext task difficulty. An extension of the training period is also needed for target dataset adaptation in the approach, predominantly improving macrospatial layout recognition while offering marginal benefits in detecting small, detail-dependent targets.

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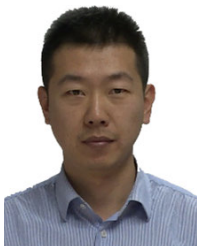
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